SQL Fundamentals : Exercise 1 Solutions

1. ```sql
   SELECT *
   FROM employees ;
   ```
   output : identical employees table.

2. ```sql
   SELECT DISTINCT department
   FROM employees ;
   ```
   output :

department
IT
HR
Finance
Marketing

3. ```sql
   SELECT first_name, last_name, salary
   FROM employees
   ORDER BY salary DESC ;
   ```
   output :

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000
   ```

4. SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 5 ;

output:

id	first_name	last_name	salary
6	Emily	Davis	62 000
3	Mike	Johnson	60 000
7	Robert	Wilson	59 000
1	John	Doe	55 000
4	Sarah	Brown	53 000

5. SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT' ;

output:

id	first_name	last_name	department
1	John	Doe	IT
4	Sarah	Brown	IT
6	Emily	Davis	IT
10	Laura	Hall	IT

6. SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 58 000 ;

output:

id	first_name	last_name	department	salary
3	Mike	Johnson	Finance	60 000
7	Robert	Wilson	Finance	59000

7. SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing'

output:

id	first_name	last_name	department
2	Jane	Smith	HR
5	David	White	Marketing
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing

8. SELECT id, firstname, last_name, department
FROM employees
WHERE department != 'IT'

output:

id	first_name	last_name	department
2	Jane	Smith	HR
3	Mike	Johnson	Finance
5	David	White	Marketing
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing

9. SELECT id, first_name, last_name,
department
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');

output:

id	first_name	last_name	department
1	John	Doe	IT
2	Jane	Smith	HR
3	Mike	Johnson	Finance
4	Sarah	Brown	IT
6	Emily	Davis	IT
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
10	Laura	Hall	IT

10. SELECT *
FROM employees
WHERE department = 'IT' AND salary > 50 000 AND city = 'New York';

output:

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

11. SELECT first_name, lastname, department, salary
FROM employees
WRERE (department = 'Finance' OR department = 'Marketing')
AND salary > 52 000
ORDER BY salary DESC;

output:

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53000

12- SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');

output:

city
Los Angeles
San Francisco
Houston
Chicago

13. SELECT first_name , last_name , department,
salary , hire_date
WHERE department != 'Finance'
AND salary > 50 000
ORDER BY hire_date ASC ;

output:

first_name	last_name	department	salary	hire_date
Emily	Davis	IT	62 000	2015-02-14
David	White	Marketing	52 000	2016-04-10
Jessica	Moore	HR	51 000	2018-05-22
John	Doe	IT	55 000	2018-06-15
Sarah	Brown	IT	53 000	2021-03-25
Daniel	Clark	Marketing	53 000	2022-06-01

14. SELECT first_name, last_name, department, city
FROM employees
WHERE city IN ('chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3 ;

output:

first_name	last_name	department	city
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago